Flow analysis of a Micromixer in Liquid Flow Cell for Transmission Electron Microscope based Study.

MECH5845M Professional Project:
Flow analysis of a Micromixer in Liquid Flow Cell for Transmission Electron Microscope based Study.
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MECH5845M Professional Project

TITLE OF PROJECT

Flow analysis of a Micromixer in Liquid Flow Cell for Transmission Electron Microscope based Study.

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This project report presents my own work and does not contain any unacknowledged work from any other sources.

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Date: 11/08/2022

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ABSTRACT

In recent years, the use of microfluidic devices for the applications of particle synthesis has been highly increasing(Douglas J. Pike, 2013). Primarily inorganic, metallic and ploimeric particles with various functionalities and material characteristics are studied(Douglas J. Pike, 2013). These Microfluidic devices have shown considerable impacts on biomedical diagnostics and drug development, while extensive amount of application are observed in the food and chemical industries(Chia Lee et al., 2011).

Homogeneouty is requirement to achieve a mixture upon which spectrographic studies can be accurately performed. For this project, mixing methodologies are designed to generate better mix of fluid and obtain clear and smooth viewing for image spectroscopy and particle tracking study in TEM(Transmission Electron Microscope).

This project takes help of computational fluid dynamics to investigate in the flow pattern and discover effect of geometry parameter changes. Multiple designs are modelled and analysed to evaluate mixing quality and flow pattern data to optimize the device performance. Also, there is a limited data available of the flow pattern in microfluidic system, Further information and investigative data is generated through the help of this project.

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Chapter 1. Introduction

1.1 Introduction

Currently in the field of fluid dynamics it is necessary to develop methods to control and handle the flow for the spectroscopic studies. Specifically, exploration of fluid flow at micro scales have just recently been around and new sub-subject are emerging that deal with fluid interactions at microscopic scales. Biochemical analysis studies like drug-delivery, protein folding, drug-tracking operated in SEM (Scanning Electron Microscope) & TEM, assert that a speedy mixing of multiple solute and substrates is essential before any chemical reaction begins (Nguyen et. al., 2005). Especially, the studies associated with TEM spectroscopy(e.g. viral particle analyzing studies performed by Janosh Houser)(Hauser et al., 2020) which has high importance in biotechnology, drug development, food analysis and Material Characterisation studies.

Microfluidic devices are developed and they have provided significant change in the industries in terms of surface to volume ration, sample consumption and low-cost systems. Therefore it is essential to develop a testing instrument of smaller scale for such studies as, experimental studies usually require proper homogeneously mixed soultion with the solute particles(for e.g. enzymes) (Williams M, et. Al. 2008). It is crucial to know the influence of study time and mixer length to achieve a processable study sample. Therefore manufacturing and designing of micromixers and flow cells is becoming more and more popular.

Currently, companies like Microfluidic Chip-Shop are leading in chip manufacturing process with various chip designs like cross-channel, straight, obstructing chips for studies like electrophoresis, spectroscopy, and micro-mixers (Elve flow, 2022). Current development in liquid flow cell holders is led by Hummingbird Scientific, The Hummingbird Scientific took the initiative of manufacturing and commercializing liquid flow cell holders for the application of liquid-electrochemical tests conducted inside TEM (Hummingbird, 2015).

1.2 Aim

The aim of this project was to investigate the liquid flow cell design of the Hummingbird Scientific and understand the internal flow structure. Further, to generate thorough and rapid mixing of Multi-species, continuous, microfluidic flow inside liquid flow cell for the application of image spectroscopy study in TEM (Transmission Electron Microscope).

1.3 Objective

- Reviewing Literature Review and Background Research of the Flow Cell and Flow Mixers for Microfluidic flow systems.
- Defining the requirements, properties, and functions of the flow cell device currently in use and discussing it with project supervisor and associated people working in this field.
- Learning analysis software like COMSOL Multiphysics and developing Geometric model of the base design(provided by Hummingbird Scientific) flow cell using COMSOL drafting tools.
- Developing a micromixer design to enhance mixing in the low reynolds number flow inside the microchannel of base design.
- Running a stationary and time-dependent study for the preliminary base and updated designs to generate output plots for result visulaisation.
- Testing of modified computational model using the amelioratory design on the current flow cell to analyse the performance.
- Critically analysing the results and developing changes to improve results.
- Reoptimizing the design and suggesting the new design changes which will improve the device performance with robust results. Finalizing the design to be presented.

1.4 Report Layout

- Chapter-1 Introduction: Initiation and overview of the project and description of the project with aim and objectives.
- Chapter-2 Literature review: Summarizes the theoretical principles behind microfluidics and flow cell is the main focus of this section. It provides overview of the flow functioning and past works and designs by other researchs.
- Chapter-3 Implementation of CFD: this section refers to why CFD study is important and functioning overline of the computational process.
- Chapter-4 Study on Base Design: This section provides details of base design study and the generated result through the analysis, as well as discussion on possible geometrical modification.
- Chapter- 5 Optimized Model Study: This section is where micromixer designs are selected and multiple alterations are processed to obtain the best design.
- Chapter- 6 Conclusion: Conclusion of the project with discussion and future work aspects are discussed in this last section.

Chapter 2. Literature review

2.1 Introduction

The research in Microfluidics is gaining popularity because of develpements in the field of microbiology and the rise in manufacturing of microminiaturized devices(Elveflow , 2022). The literature describes the general idea of microfluidics, flow cells and their flow properties. New emerging mixing technologies in the research field are discussed for which the relevence of flow properties and equation governing the flow shows relevence.

2.2 Microfluidics

Recent developments in the field of drug development and microbiology were possible because of improvement in microfluidic testing devices, creating opportunities for development in microchannel flow, nanofluid sensor (Elveflow, 2022). Any flow devices where the channel dimensions are smaller than millimetres (usually Micrometer scale) are referred to as microfluidic devices. These dimensions can go down to the scale of nanometres also. Similarly, microfluidic flows refer to flows that have a continious supply of fluid inside a channel of micrometer size range. The biggest advantage of these kind of devices as micromixers are that they can control the number of droplets, concentration of the chemicals and the sorting of liquids (Y. C. Tan, et al. 2004).

2.1.1 Flow Cell

A flow cell is a device that facilitates a flow transport and mixing of continuous microscopic fluid flow in a specific path (Hummingbird , 2022). In these devices the new samples are constantly replaced to ensure that there is no kind of interruption, clogging and distortion occurring with the flow. A specific type of flow cell holder, designed by the hummingbird Scientific, promotes continuous flow inside a mixing chamber with the function of keeping the specimen in a liquid environment and protecting it from contamination and damage (Hummingbird, 2015). This device also allows multiple flow mixing at the tip of the device and assist liquid flowing at same or different rates (Hummingbird, 2015). For Spectroscopic studies of the specimen this holder has Energy Dispersive Spectroscopy (EDS) option that is an elemental analysis

Figure 1: Hummingbird Scientific's TEM liquid flow holder(Hummingbird Scientific, 2022)

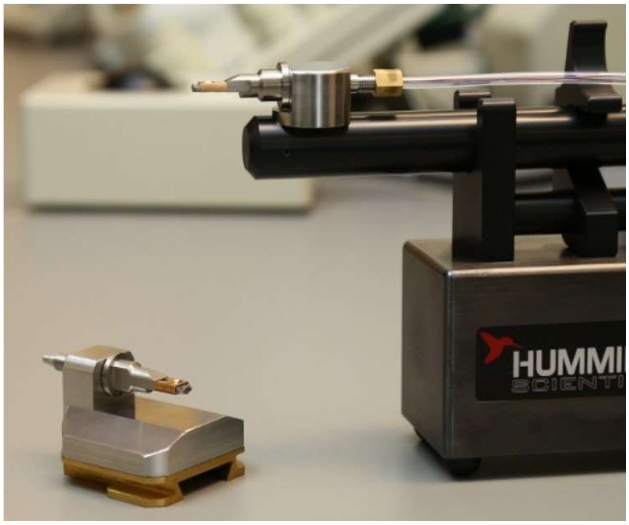

Figure 2: Hummingbird Scientific's SEM liquid cell(Hummingbird Scientific, 2022).

Technique for element characterization studies. These types of holders designed by hummingbird Scientific, these devices are especially capable of producing the thinnest possible liquid layer at the viewing window, allowing researchers to perform electron energy loss spectroscopy study (EELS) and perform elemental analysis and mapping at a very high-resolution magnification (Hummingbird, 2015).

2.1.2 Flow Properties

2.1.2.1 Laminar flow:

The flow occurring in a flow cell is on a microscopic scale; thus, the flow occurring has a low Reynolds number, and for such kind of flow, no turbulence is to be discovered. In such a case, microscale mixing is necessary (Douglas Pike et al., 2013). As for macroscopic mixing, a turbulent component in the flow is a requirement. It is physically impossible to generate turbulent flow for such a low Reynolds number and mixing in laminar flow needs to be done. In such laminar flows, the advection (flow transport in direction of flow) component is extremely low, and most of the mixing of the two liquid flows occurs due to diffusion (governed by concentration gradients inside the flow) (Douglas Pike et al., 2013).

One such flow condition is assumed by one research study where a steady-state flow has a constant flow velocity of fluids and different solute concentrations that are mixing in a T-shaped channel, and appropriate mixing occurs at the boundary of the two fluids (Park et al., 2017). Here, to determine the status of the flow, A Reynolds number calcuation can be used. To reach the appropriate concentration of the influent after mixing, a precise flow pattern is required as it determines the flow rate and flow volume (Douglas Pike et al., 2013). In some regions, eddies are developed where the channel width is high, and these patterns inside the flow cells determine how fast the concentration of analyte fluid reaches the influent [injected fluid] (Douglas Pike et al., 2013). The flow pattern changes with the change in flow geometry and flows velocity,

and this change needs to be studied for different mixer geometry and flow rates (Douglas Pike et al., 2013).

2.1.2.2 Reynolds Number:

In microfluidics, because the channel width is in the scale of microns, most of the fluid flows have laminar flow. A key factor upon which the flow is characterized is the Reynolds number.

$$Re = \frac{\rho u D}{\mu}$$

Where: ρ= Density of the fluid

u= Velocity of the fluid

D= Diameter of pipe

μ= Dynamic Viscosity of the fluid

If a flow has a Reynolds number value lower than 2300 it is laminar and if more than 4000 it is turbulent. For the scenario where Reynolds number is in between 2300 and 4000, the flow is characterized as transitional (Cengel and Cimbala, 2006). In Microfluidics, even though the flow has a higher flow velocity, the Reynolds number can go as low as 0.01. Although a variety of different flow rates can be used for the study of devices like these but flow rate in range of 10 to 0.1 ml/min were used (Douglas J. Pike, 2013). Therefore, the Reynolds number generated was in the range of 100 to 0.5.

2.1.2.3 Peclet Number:

Determination of the mixer functional range is characterized by the reynolds number of the flow. However, Peclet number is a dimensionless quantity that helps in detrmination of the transport of the solutes inside the flow(Prichard et al., 2022). In COMSOL Multiphysics by solving the convective diffusion equation(commonly known as advection diffusion equation) we can generate peclet number value.

$$P_e = \frac{ud}{D}$$

Where: u= Velocity of the fluid

d= Diameter of pipe

D= Diffusion Coefficient

The Peclet number can be used to determine the minimum mixing length for optimal mixing quality.

2.3. Mixing Methodologies:

Numerous studies present schemes by providing flow samples through multi-channels, holes and microchannels with introduction of obstacles. The main two methods that are discussed are stated as follows:

2.3.1 Active Mixing Methodologies:

Extensive number of active methods are introduced in newer microfluidics devices for increasing mixing performances. Active mixing methods are driven by the principle of exciting and addition of chaos into the fluid flow by introducing external energy to the flow (Chia-Yen Lee et. Al. 2016). Generally, this method is implemented by the incorporation of mechanical actuation and generally has a moving component. Some of the more promising approaches are of using acoustic, dielectrophoretic (DEP) force actuation, magneto-hydrodynamic flow, electrochemical instability to advance in stirring of the fluid flow. Figure 3 shows DEP micromixer with formation of circulating transverse flow by AC electroosmosis for flow mixing (M. Campisi, 2007).

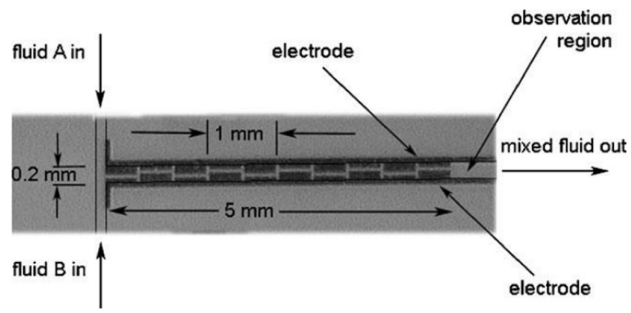

Figure 3: T-channel design by M. Campisi et. al. for DEP micromixer design(M. Campisi, 2007).

2.3.2. Passive Mixing Methodologies:

Passive mixing methods are more established and the applications are easier where external influence is not

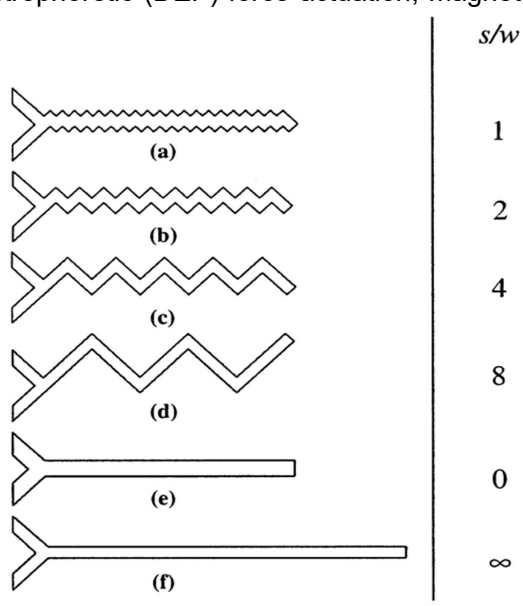

Figure 4: Geometries deigns by Mengeaud of the microchannel with different periodic step to width ration(Mengeaud et. al. 2002).

desired. The passive mixing methods are one where there is no moving part present, and no external energy source is used to introduce additional turbulence. An approach of increasing the contact is of mixing fluids so that fluids interact at multiple points. Another factor that assists in mixing performance by increase of contact time of the liquids (Chia-Yen Lee et. al. 2016). The mixing in a passive mixing relies on chaotic interactions by an increase in contact area and time. Serpentine structure, T channel, Y channel and lamination-based micromixers are the most efficient designs which provides optimal mixing of the fluids with only the help of geometric modifications. Figure 4 shows Mengeaud et., al.'s zigzag based passive mixing channel which was found to be extremely efficient in creating turbulence in vertical directions, hence providing better mixing of flow at the outlet even for lower inlet flow rate (Mengeaud et al, 2002).

2.4 Numerical Study

2.4.1 Mixing Analysis:

To obtain a homogeneous mixture of fluid concentration, design variation and addtition of mixromixers is key. Various quantity and profiles can be used to determine mixing. However most easiest and effective means of mixing analysis is by concentration. Variation in concentration at specific location with the average concentration inside the flow system provides information about the quality of mixing.

2.4.2 Mixing Quality Evaluation:

A concentration field shows the varied concentration density throughout the evaluation area; although, to calculate how well fluid is is the didtribution of the concentration in the micromixer, Mixing index is to be calculated. The mathematical representation for mixing index across a specific area is provided by the equation shown below:

$$\sigma_1{}^2 = \int_A^n [\mathrm{c1} - 0.5]^2 \mathrm{u} \cdot \mathrm{dA} / \int u \cdot dA$$

$$MI = 1 - \sqrt{\frac{\sigma_1}{\sigma_2}}$$

where, MI = Mixing Index

σ = Concentration variance

c = concentration value

u = flow velocity

The variance equation here is for the flow system with a single entry point therefore the concentration of the system will be highest 1 and minimum 0 thus average concentration will be 0.5.

Chapter 3. Implementation Of Computational Fluid Dynamics

3.1. Introduction

From the above section, we can firmly say that significance of Computational Fluid Dynamics (CFD) is essential to analyse the flow pattern and generate flow data. Quantitative approaches based in laboratories are performed by use of florescent dyes into the solute to distinctly visualize the flow pattern and concentration gradient.

CFD is computational methodology that works upon the physical principles and mathematical calculations to generate numerical result of fluid principles and interactions (Thabet, et al. 2018). Generally, the fluid flow studies are approximation of the study performed in experimentally in the labs; however, using CFD is more efficient as we for saving resources, time, and produced multiple results. Therefore, a proper model of study needs to be made to generate the most accurate results. For the current study, COMSOL Multiphysics is used as it is the most reliable software to perform Multiphysics and parametric study. The whole study can be divided into multiple sections which are as follows.

3.2. Model Selection

From the earlier discussion we know the flow properties and the output variables that are needed to calculate. Therefore, proper model selection needs to be done in order to generate the required solution that needs to be decided. Concentration field is a variable which can be calculated by calculating the movement of specific particles throughout the analysis. Transport of diluted species model is the perfect method to calculate particle movement.

All the constrains in flow properties make the flow occur in a specific property domain where the flow will be extremely laminar with low Reynolds number therefore laminar flow model of COMSOL is suitable to project laminar conditions to the flow. Finally, the integration of both models is required and COMSOL provides a facility to couple both

the models into one study. In response, a reacting flow, diluted species multiphysics was set to simulate mass transport and flow variation in liquids.

3.3. Solver

Factors of this study like performance, time, and accuracy highly depend on solver configuration. Numerous computational solvers are available for different software tools and selection of solver varies with the study problem. COMSOL multiphysics provides two types of solvers namely direct and iterative solvers. Although direct solvers, like PARDISO, are capable of generating solution as accurate as iterative solvers, they come with certain disadvantages.

Direct solvers provide the solution by working over one huge step which causes high memory usage (Walter Frei, 2013). Iterative solvers take the upperhand in terms of computational time and error estimation. Although they have higher risk of failed study(Walter Frei, 2013). If the boundary conditions are well defined, iterative solvers will give accurate and faster results. Another advantage is the error estimation for each iterative steps can be analysed by an error convergence plot which informs if the problem is set correctly or not. Considering all these factors GMRES iterative solver was selected for the study.

3.4 Governing equations

3.4.1 Advection Diffusion Equation:

When a flow occurs through a microchannel, the force that drives the flow is dependent on the equation known as the Advection Diffusion equation. Numerically, a Time-dependent advection-diffusion equation is studied to obtain the flow pattern for a laminar incompressible flow.

$$\frac{\partial c}{\partial \mathrm{t}} = D\nabla^2 c - v \cdot \nabla c$$

Here:

$D\nabla^2 c$ is the diffusion component,

$v \cdot \nabla c$ is the advective component.

A diffusion driven transport is controlled by this equation and concentration gradient is obtained. The response of flow is dominated by diffusion because of low reynolds number. The advective component can be increased for more turbulence by implementing mixing methods.

3.4.2 Explain Navier Stokes Equation:

The solution for the flow fields were generated by solving 3D Navier stokes equation for incompressible flow. Navier stokes equation can be solved to determine laminar variables and obtain reynolds number data. Numerical interpretation is provided below,

That is:

$$\rho v \cdot \nabla v = -\nabla p + \mu \nabla 2v \qquad \& \qquad \nabla \cdot v = 0$$

Here: $\rho v \cdot \nabla v$ = Acceleration Component

$-\nabla p$ = The pressure gradient

$\mu \nabla 2v$ = Viscosity Component

3.5. Post Processing

The last step in a CFD study is post processing, this contributes to analysis and visualization of the results. Along with Ansys, OpenFoam, etc, COMSOL Multiphysics has high postprocessing tools for 2D, 3D plots. An accurate calculative method for golbal variables evaluations is also provided by the software. An accurate and concise visualization and result processing is achieved by COMSOL which was displayed.

Chapter 4. Base Design Micromixer Computational Model

4.1. Introduction

Preliminary computational study was required in order to understand the flow pattern and profile. To verify the model selection, it had to be implemented on the base design which is designed by Hummingbird Scientific and currently in use.

4.2. Geometry

A geometry of the flow cell holder that is developed by hummingbird scientific is replicated in COMSOL Multiphysics with the dimensions provided by Hummingbird Scientific[Figure 5].

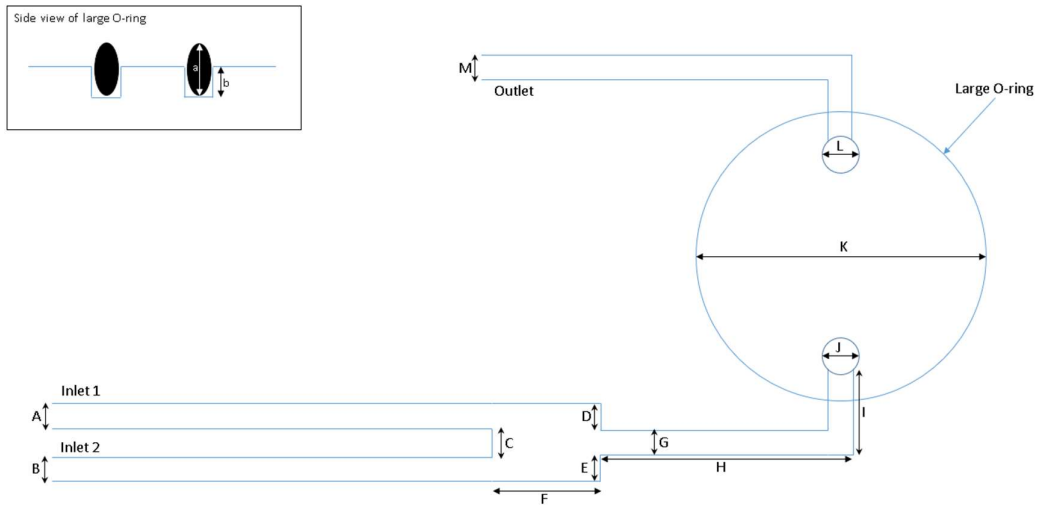

Figure 5: Schematic Design of the base geometry provided by Hummingbird Scientific.

The geometry prepared in COMSOL provides a generic idea of the structure of the flow cell holder and the dimensions taken were provided by hummingbird. Although a range of thickness was assumed for the device with both models of cylindrical and cubical shaped inlets were built (Figure 6). Here F is the mixing tube length which is 9000μm. The inlet cross sectional length is A & B(150 μm). However in base design inlets are placed at a 100 μm distance(C) from each other. The flow is vented out through the outlet channel with diameter G that is 150 μm. Further the flow reaches the viewing window for visual spectroscopy. However this project will be focused on the microchannel geometry.

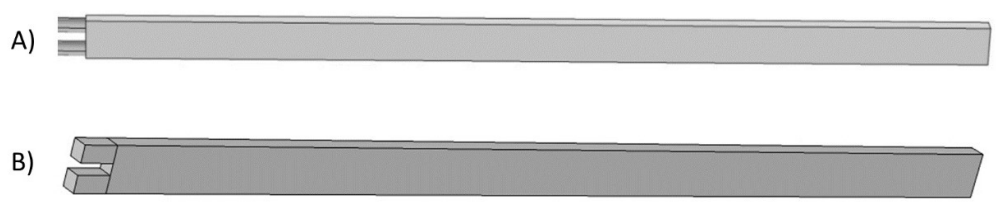

Figure 6: Geometrical design of flow cell mixing component . A) With cylindrical Inlets, B) With square Inlets.

This design is the base design upon which a flow study is performed so that flow pattern can be understood, and further optimization can be done. Essentially, the Geometry consists of two inlet components brought together into a single channel mixing chamber. From our most primitive understanding we know that when two different liquid come into contact inside the channel the mixing will be dominated by diffusion (Schwesinger et al., 1996). Here, barely any external turbulence available; therefore, it is assumed that diffusion will be the driving factor of the mixing.

4.3. Model Setup

4.3.1. Initial Boundary Condition:

At inlet the boundary condition of a fully developed flow with a flow rate of 10, 5, 1 & 0.1 ml/min were set so that Reynolds number can be properly managed and results were compared for each. Peclet number evaluation is not evaluated in this project as another approach for mixing length calculation was employed. Through the reference of work done by Liangchao Zou, 2017, fully developed flow rather than a pressure variable was employed as here we need equal flow rate from both fluid while in case of uniform pressure boundary condition different flow rates can be observed (Liangchao Zou, et al., 2017). This boundary condition was assigned to two of the inlets of the base design geometry, while at the outlet a 0-pascal pressure boundary layer was set which will direct the flow from inlet to outlet. Additionally, a no backflow constrain was implemented to avoid any pressure reluctance at the outlet and stop backflow.

The boundary surface of the geometry was assigned a wall function to give a body to the flow plus a no slip condition was set. Now it is required to generate data for concentration profile and mixing profile of both the fluids, so a transport of diluted species condition provided by COMSOL Multiphysics was selected. For the study to be even simpler the fluid 2 concentration was kept at 0 and fluid 1(water in this case) at 1 so that the flow concentration profile can be visualized with fluid 1.

Here both stationery and time dependent study were performed and the initial concentration of 0 at time = 0 was used. A relative convergence plot criteria was set at 0.0001 for the stationary studies where solution id provided by GRMES solver. Because the model geometry here is in 3D and working on COMSOL which is a finite element software, it takes a huge amount of computational power, now there is not much modification that can be done into the geometry so that the computation will be faster. Although into the study steps the laminar and transport of diluted species study were done into two separate studies to boost up the computation time. Another method that was implemented was that dividing the geometry into smaller section like 3 obstructions rather than 11 which improved the computing time.

4.4. Results & Validation

The primary focus on the result was to obtain a velocity profile and a concentration profile. Now, the concentration gradient that was obtained was steady for all the different flow rates.

4.4.1 Velocity Profile:

The velocity profile was generated to check if there is a good agreement with the profile with the references available through other studies. From the stationary study similar results for velocity profile were found form the results generated by computational study performed by Matthew Itomlenskis for a rectangular microchannel provided in Figure 7. (Itomlenskis et. Al. 2008).

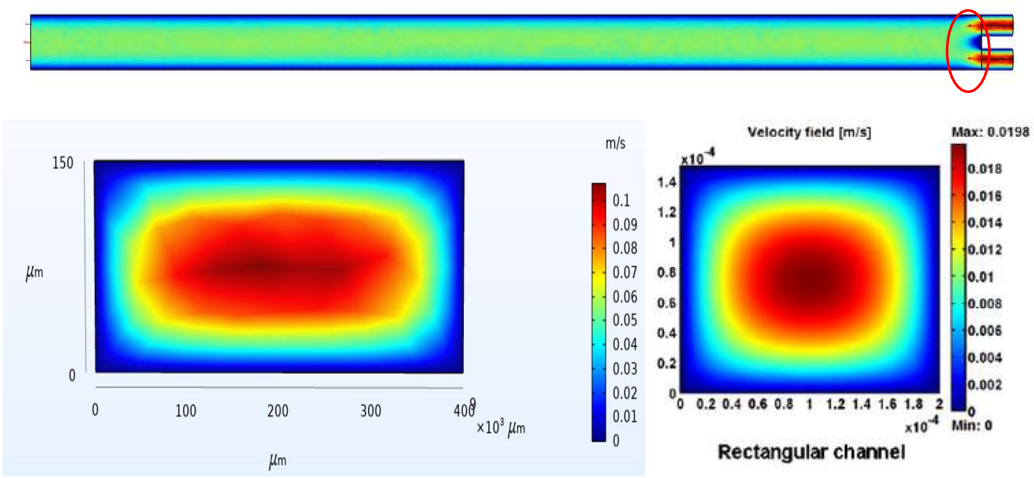

Figure 5: Velocity profile of the flow through the whole device (Top). Velocity Field at outlet of Base Design (Bottom Left) Compared to Velocity Profile in rectangular channel generated by Matthew Itomlenskis(Bottom right)(Doughlas Mike, et al., 2013).

Here it can be seen that there is variation in magnitude because the size of the channel and inlet flow velocity is not same in both the study but a proper convergence in profile is seen. The maximum velocity that was obtained at the outlet was 0.1 m/s.

Although an interesting phenomenon, similar to the experimental study of Doughlas P. Mike, was seen where in the velocity profile neared the inlet, a reversed motion of fluid was seen (Figure 6). Here it can be seen that because of the increase in channel

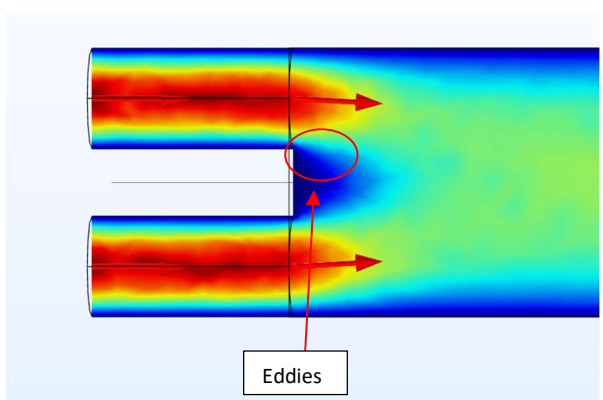

Figure 6 Velocity Profiel of base profile showing the eddies formed near inlet.

width a type of eddie formation occurs even though the flow is extremely laminar(Douglas Pike, et al. 2013).

4.4.2 Concentration Profile:

Another result of interest was the concentration gradient and concentration profile. The study works upon the transport of diluted species physics which relies on the diffusive law and advection diffusion equation to gather concentration data. The concentration profile generated shows both the fluids merging together, Although the flow rate is so low that there is barely any turbulence inside the flow. Therefore, mixing occuring is mostly dominated by diffusion. Due to these reasons, a fairly consistent intermedent concentration gardient is generated which differentiates both the flow concentrations.

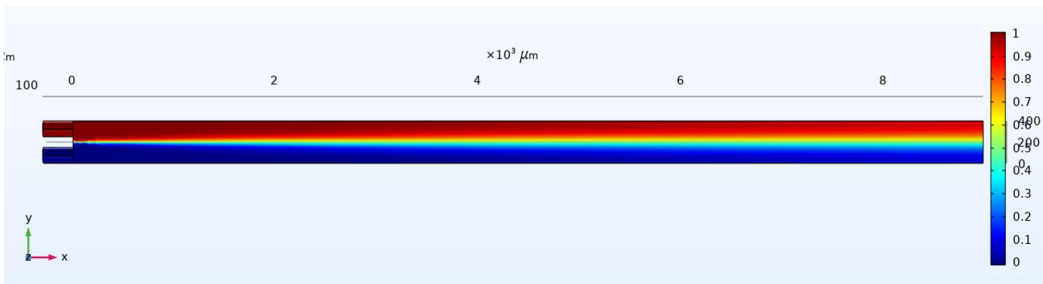

Figure 7: Concentration profile of base design showing concentration gradient growinginside the flow cell microchannel.

In the figure 7 homogeneous mixing of two fluids with an increase in the length of the channel simiar to the hearing-bone channel study is observed(Mondal Bappa et. al. 2019). From this result it can be easily assumed that the mixing obtained is not optimal and only changing the length of flow cell to an exorbient amount would improve mixing which is in practice not suitable designing approach. The purpose of the study was to define how optimal the mixing of the double flow in the channel is thus a quantity named Mixing Index(MI) was calculated. Now the mixing index for two designs one with sqare inlet and with cylindrical shape inlets were 0.409 and 0.419, respectively. The Figure 8 shows the mixing profile at outlet. As the flow proceeds through the length the mixing gets better. So increasing the length of the mixing chamber can increase mixing index but that is not an optimal method as it will increase the size of the apparatus. For an optimal design, high mixing index with lesser micromixer length is desireable.

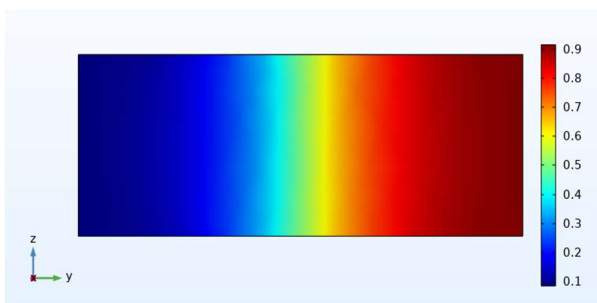

Figure 8: Concentration profile at outlet cross section showing the concentration distribution after mixing.

4.4.3 Mesh Dependency test:

Software packages like COMSOL Multiphysics uses a finite element method to solve the flow equations, and to find the results a mesh convergence test needs to be done. However, due to the method of operation of the software, the result of the concentration gradient is dependent on number of mesh element and element size. The mixing value depends on dependent variables that is concentration thus even though making the mesh finer there would barely be any convergence. The reason for the mesh convergence test to fail is that the mixing index that is plotted with number of mesh elements is a dependent variable, concentration depends highly on mesh fineness. Therefore, to select the most optimal mesh a convergence of velocity with respect to the number of degrees of freedom was calculated and associated to that an optimal mesh selection was chosen as shown in Figure 9 below.

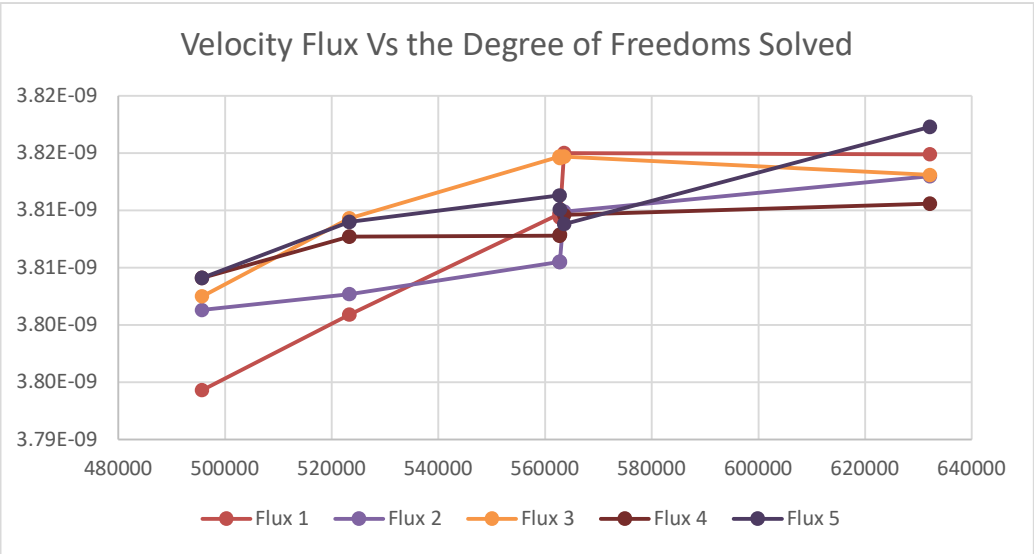

Figure 9: Convergance plot of velocity flux at outlet with respect to different number of DOF solved by GMRES solver.

4.4.4 Spected geometrical changes

To increase the performance of the devices an understanding needs to be build where geometrical changes will drive the performance. Some of the suggested changes to understand flow pattern and changes are:

4.4.4.1 Change in Distance Between Inlets:

To understand the flow pattern at microscales and with extremely low flow rate, A notion of geometrical change of inlet distance modification was implemented. In the current design the distance between the inlets is 100 µm, even though it was suspected that by placing the inlets closer there might be formation of advective eddies formation on the edged where the concentration less mixed. Another suspected phenomenon

was as the flow inlets come closer the contact region of the flow will increase and the mixing gradient will become more wider which means more better mixing of fluids at smaller length of mixing chamber.

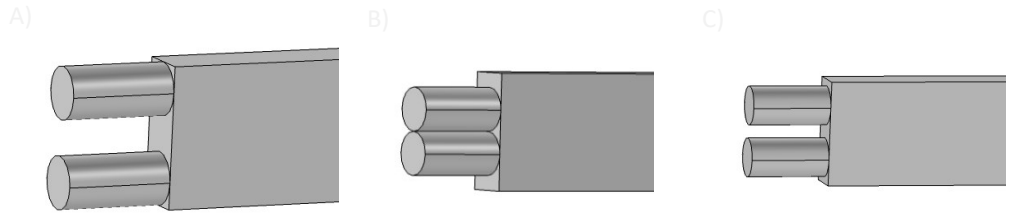

Figure 10: Image showing geometric modification at inlet where distance between inlets values are A) 100µm, B) 0 µm, C) 50 µm

Therefore, to verify these speculations a stationary study with same boundary condition was performed. In this case, 7 models were built with distance between inlets varying from 100 µm to 0 µm, three of them are required to visualise the variation in result.

Now similar concentration profile and velocity profile with pressure drop is generated for all the designs.

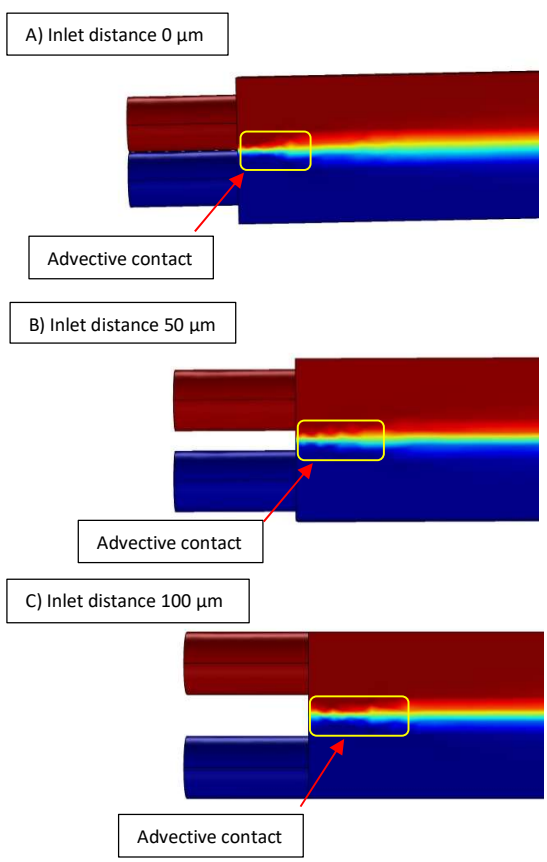

Figure 11: Concentration profile showing advective contact for different changes in base design.

The Concentration Profile and Velocity Profile of all the geometry is generated. However we need only three of the designs to get a jist of concentration gradient. For reference, concentration profile of three geometry is displayed in the Figure 11.

From the Concentration profile it can be seen that there is a disparity between the turbulent contact length of the two flows, a least contact length is seen (Figure 11) where the inlet length is the least while which was not speculated, the reason is the lower flow rate as there is not enough advective force in the flow to generate chaotic advection thus the flow disperses with diffusive mixing.

To understand this, a graph plot of Mixing Index versus distance between inlet is plotted. Here it was discovered that maximum mixing was obtained at a distance of 30 µm a curve where there is an increase in the MI with lowering of inlet distance; but after a point MI decrease again and becomes almost similar (Figure 12). Here, another unusual speculation is observed that is the phenomenon of demixing as defined by Norbert Schwesinger in his study of modular microfluid system(Schwesinger et al., 1996). Here rather than improving the mixing by external eddies at the edges cause a backflow which separates the two flow which causes a phenomenon of demixing. Therefore, graph plot shown in figure 12 is observed.

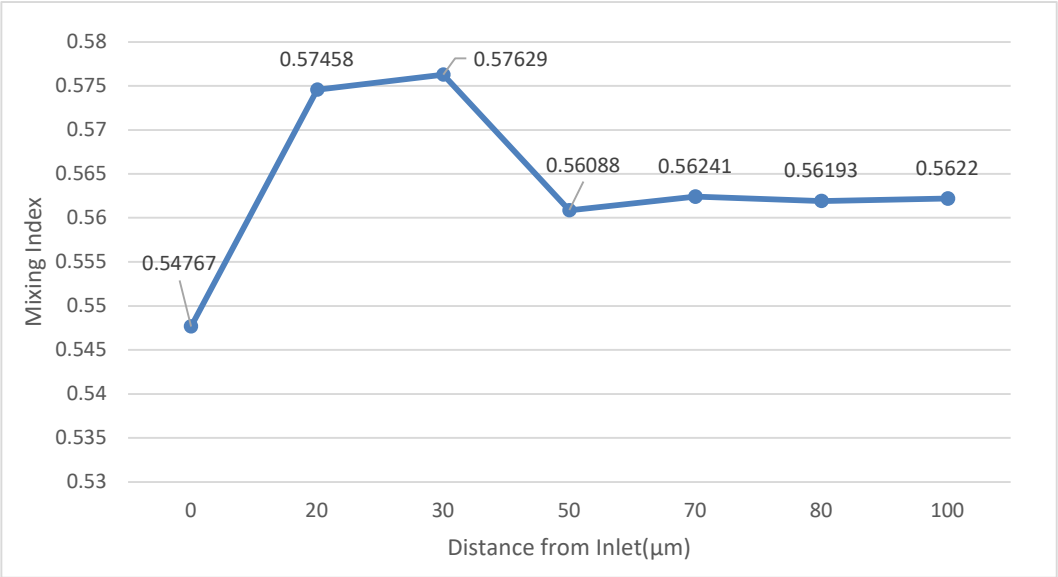

Figure 12: Graph plot for variation of mixing index with parametric change in inlet distance from each other.

4.4.4.2 Length Modification of the mixing Chamber:

Another simple change in geometry was by increasing and decreasing the length of the mixing chamber. Although a very minor change in MI was observed with even doubling the chamber length. Thus, it was concluded that simply changing the length will not significantly improve the device mixing performance.

4.6. Discussion

From the provided studies by comparison of the result of MI of each study, it was observed that change in flow rate and change in the length of the mixing chamber provide a negligible benefit for device performance. For the base geometry, the modification of the inlet channel from cylindrical to square inlet helps in improving the processing speed of the geometry with barely any change in the result. Some irregularities in the design change by changing inlet distance was observed where a phenomenon of demixing and reverse eddies are introduced into the flow mixing

causing an irregular change in MI with change in inlet distance. This result needs to be kept in mind while working on further design modifications which entails adding chaotic advection into the mixing chamber. These results are also supported by the william s. Manda's work for hearning bone channel result for eddie formation because of hearingbone channel obstruction in the flow path (Williams et al., 2008).

Chapter 5. Optimized Micromixer Computational Design

5.1. Introduction

In order to quantify the maximum performance of mixing chamber in a continuous microfluidic system. Multiple passive mixing designs were implemented to entertain chaotic advection-based microchannels. As presented in the literature review it is clear that mixing methods which induce turbulence in different directions are favourable for better mixing. Implementing those geometries into the COMSOL Multiphysics computational model is done in this section to generate appropriate results. From the two mixing methods mentioned earlier in the literature review, active mixing methods consist of methods involving interferences from external factors like introduction of, electromagnetic conduction, heat variation, energy increase of system. This kind of method is not suitable for the device hummingbird scientific is trying to build. It interacts with the working of the flow and causes irregularities in the functionality of the device. Now a passive mixing method is preferred because there is only geometric modification that helps in introducing advection to the flow. Many CFD studies are developed recently with passive mixing chambers, For Example in comparision data provided by the work of Chia-Yen Lee, some prefered designs were selected to implement on the base design that is to be discussed(Chia Lee et. Al. 2011).

5.2. Design Selection & Implementation

From the literature review section, by discussion regarding micromixer designs, two designs were selected, which are simulated in COMSOL Multiphysics software. From the study results of base design, it was concluded that a vertical component of the velocity is required in order to obtain a homogeneous mixture of the fluids. To overcome this issue, design with barriers that induce vertical and horizontal advection is introduced. Design study for each of these designs were build that are elaborated below.

5.2.1. Design 1(Advective obstructions)

5.2.1.1. Geometry

A standard geometry provides an easier computational model, although modification in the design is required. For the first scenario an array of obstacle blocks were introduced into the mixing chamber. Obstacle shape, number of obstacles, and positioning of obstacles were selected by trial-and-error method. Two designs with circular and square obstructions were designed. To save memory of computing device a preliminary geometry of array of 3x3 obstacles were created for a mixing tube length of 800 µm. All other dimensions were same to that of the base design. Figure 13 shows the geometry of both the models.

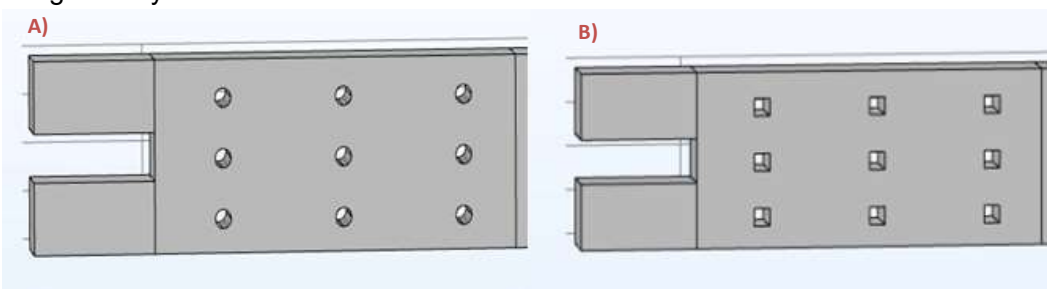

Figure 13: Preliminary design 1 geometries. A) Cylindrical obstructions, B)Square obstacles

5.2.1.2. Model Setup

Initial Boundary Condition:

Similar boundary condition to that of earlier design were implemented with single species flowing through the mixing tube from two inlets and one outlet. For flow rate, as determined earlier 0.1 ml/min value was considered. Multiple integration surfaces were created throughout the geometry where flow variable values can be calculated to determine variation of dependent variables throughout the arc length.

In this study, three conditions were selected: stationary, parametric-sweep and time dependent study. The parametric sweep study helped in projecting results for different mesh quality in the same study. The variation of mixing quality with time variable is an objective which can be generated by a time-dependent study. The time range of 0 to 0.5 seconds was selected with a time step range of 0.1 seconds was set. The study results were solved by implementing a GMRES solver and the residual error limit of 0.01 was taken.

5.3.1.4. Result Obtained

Primary focus of the results was upon the velocity profile, concentration gradient, and MI value.

Velocity profile:

The velocity profile provides the flux path of the velocity of flow the velocity flux of both the geometry is represented in the figure 14. Profile shows that there is variation in magnitude because the size of the channel and inlet flow velocity is not same in both the study but a proper convergence in profile is seen. The maximum velocity that was obtained at the outlet was 0.18m/s. this design shows that eddies formation in cubicle

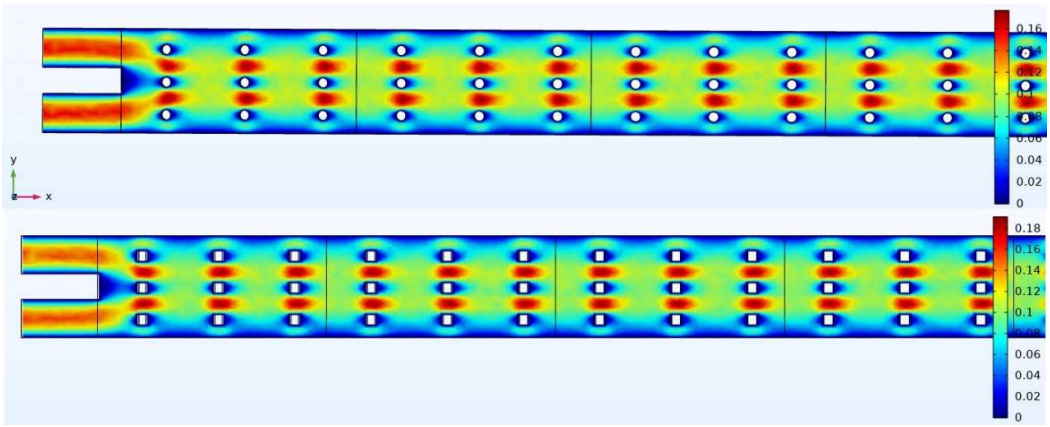

Figure 14: Velocity Magnitude profile of design 1 by stationary study.

obstacle design is higher than circular design. Therefore more dissipation of flow path occurs, hence better mixing

Concentratuion Profile:

Concentration map shows the concentration at the outlet of mixingchamber (Figure 15), Specifically, at the back of each obstacle it was seen that eddie formation was occuring and therefore when two fluids separated by obstacles they tend to come diffuse with each other behind the obstacle. A mixing index for 3x3 obstacle array of cylindrical obstacle with 11 units was calculated and highest mixing of 47%(0.471). Similarly for square obstacle a mixing index of 51% (0.513) was obtained. From the concentration Profile it can be seen that

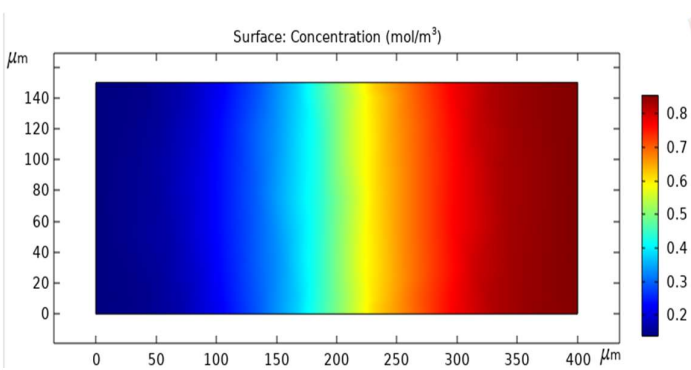

Figure 15: Concentration map at outlet, A)Circular obstacle design, B) Square obstacle design

Position and number of obstacles also affects the mixing quality of the flow to test that a mixer design with 4x3 array of obstacle was built and stationary study was conducted for same boundary conditions. Normal assumption would be that increasing the obstacles will increase turbulence in the flow. However form the concentration plot from

Figure 14, It was seen that rather than increasing mixing quality it decreased the mixing quality. The reason being a flow separation occurring due to obstacle placement which leads to the phenomenon of demixing.

Angle of Obstacle:

For the flow to occur some kind of energy is required to drive the flow, in a microchanneled flow the driving force of the flow is created by the pressure drop between inlet and outlet. Although an inverse relation is seen between pressure drops and mixing efficiency. Both quantity needs to be improved for the design to be considered good. By changing the angle of the barriers relative to the flow path, the pressure drop can be reduced. Therefore, comparison study of variation in mixing index and pressure drop with respect to the angle of the obstruction was generated as shown in Figure 16. The angle range was taken from 0 to 45 degree because the shape of obstacle is symmetric which means same results will be from 45 to 90 degree range. From the graph it is clear that keeping the square obstacle at a 45-degree angle improves the device performance by a huge aspect and that was taken as optimal angle for squared obstacle.

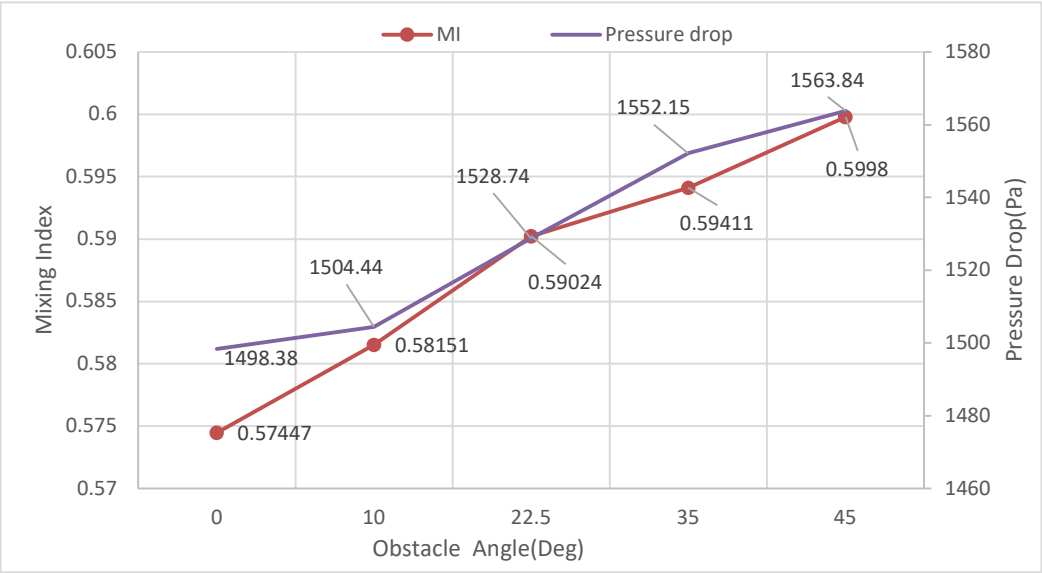

Figure 16: Graph representing Variation of Pressure drop and MI because of change in angle of obstacle.

Geometrical Modifications:

The problem with adding barriers in a uniform pattern with equalized gap pattern gives separate concentration near the edge of the mixing chamber. Better mix of fluids will occur if advective force are formed such that fluid near the edge is forced to be push towards the middle of mixing chamber. By keeping this in mind, a geometrical modification was suggested where five rectangular shaped barriers at an angle range of of 45 to 75 degree. The placement and size of barriers inside mixing chamber fabricates a motion of fluid that makes flow to split towards the edge and later makes the flow redirects towards the middle of the chamber causing chaotic advection in the channel. This type of design was selected by iterative geometrical change method. The geometry of the modification with single mixing unit is provided in the Figure 18.

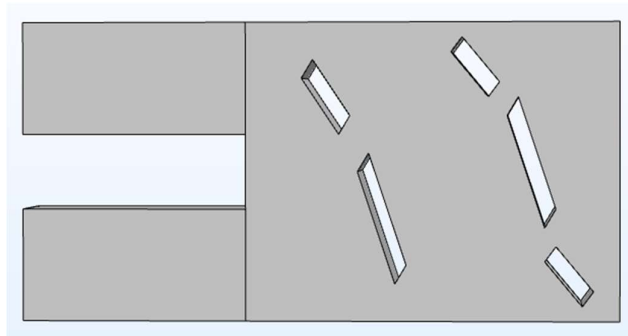

Figure 18: Design 1 with geometrical modification.

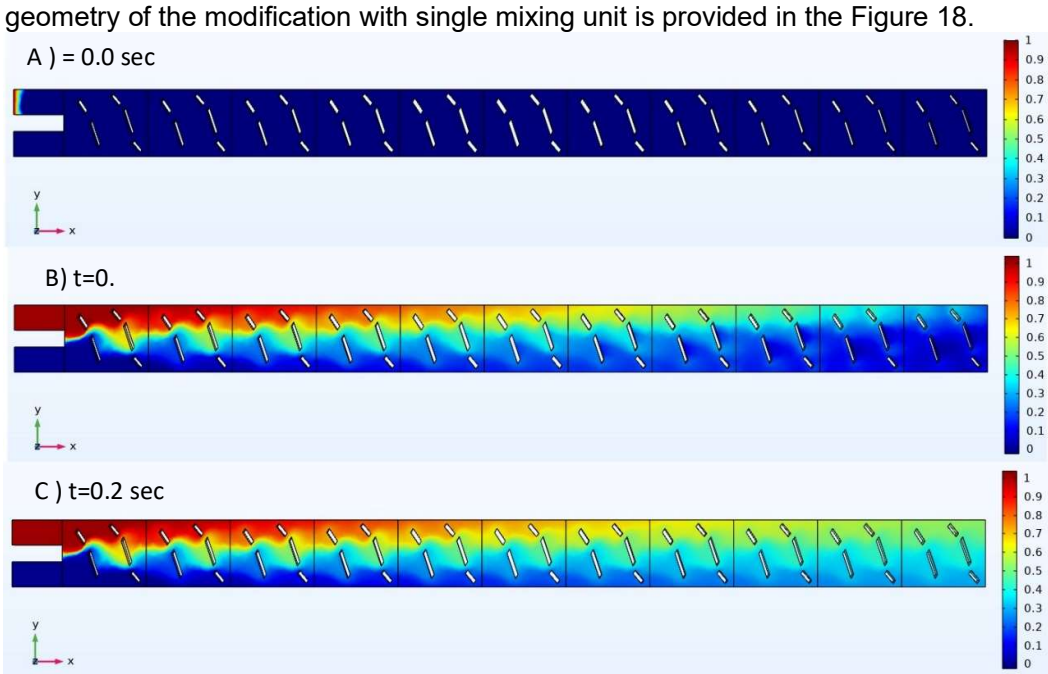

Figure 17: Concentration profile of modified Design 1, at time step A) 0, B) 0.1, C) 0.2 sec.

The concentration plot for the following geometry is developed, now 11 mixing unit were combined. The concentration map at different time step is provided in Figure 20. From the image, clear and uniform mixing of both fluid throughout the mixing channel areas. Mixing quality at the outlet significantly improved with MI value of 0.78 which was achieved from a stationary study compared to the base design.

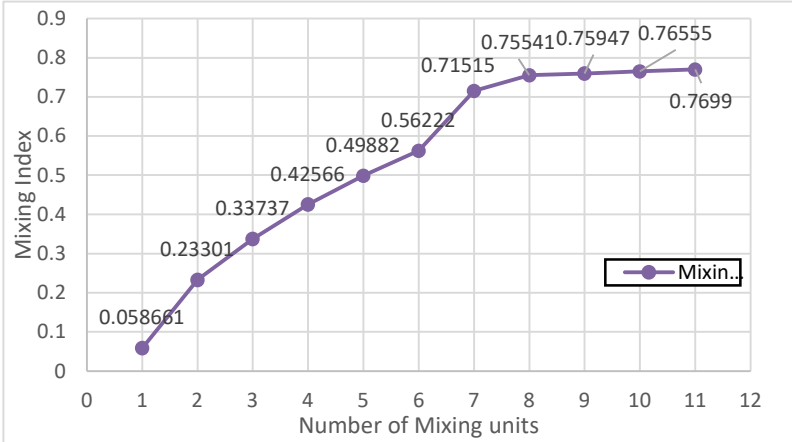

Figure 19: Convergance plot of MI with Increase of Mixing Units.

To optimize length of the mixing chamber, number of mixing units, which provides appropriate mixing value, is calculated. For this calculation a plot of number of mixing units versus the mixing index was calculated. The Mixing Index value converges even after adding more mixing units after 8 units, thus mixing chamber consisting of 8 mixing unit was selected as most optimal design.

Mesh Dependency Test:

The dependency upon the mesh is a factor in COMSOL that causes the change in output dependent variable with the grid density. To mitigate this dependency issue, an alternative method can be used where the convergence of velocity flux with the DOF solved by the software can be plotted. The mesh dependence test graph is attached in the Appendix (Section 5.3.2). Another study of variation of concentration throughout the mixing chamber with 7 different mesh sizes.

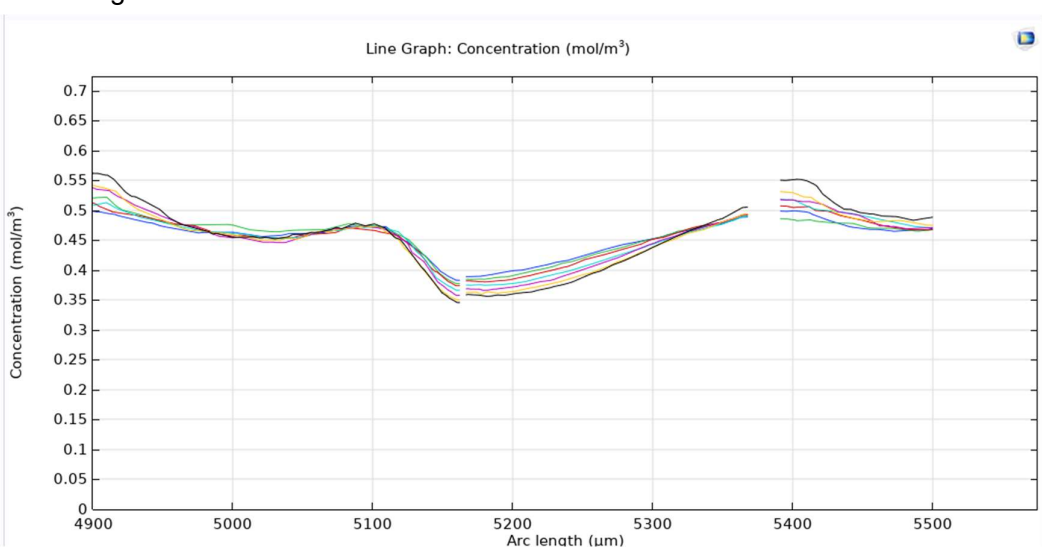

Figure 20: Concentration variation with arc length for 7 meshes is plotted for design 1.

The figure 20 shows the concentration convergence plot. Certain points in the graph shows the elements of no concentration whose are the geometry obstructions. The figure provided is zoomed of the last mixing unit for better understanding of the graph.

The graph shows the seven meshes follows same curve although some magnitude variation. At some regions the lines are superimposing means similar results are obtained for different mesh size which makes the mesh independence study sucessful. However because the geometry is complex and there is a limit to the gird size a computer can achieve further refinement of mesh was difficult.

5.3.3. Design 2 (Parallelogramic barriers)

5.3.3.1. Geometry

An approach to passive mixing, (Fang et. al, 2012) suggested a design which uses a parallelogramic barrier into the mixing chamber although the dimensional parameters taken were reformed for better results. The design of mixing chamber with use of parallelogram barriers in order to provide a zigzag flow path increases mixing in vertical direction. For that reason a geometrical design with multiple ridges was created on COMSOL Multiphysics. The geometry consists of 11 pair of barriers at an angle of 80.9 degree inside the mixing chamber that was selected as primary geometry. The geometry had a gap of 400 microns between two barriers. The geometric design was selected by multiple iterations of the study where the most optimal design was selected. For example, the iteration of the angle of barrier is discussed in later sections(5.3.4). The thickness of the barrier is taken as 20 μm so that low quality of mesh can be avoided and provide ease to manufacturing of the geometry. The geometry was similar to the base geometry with two inlets and one outlet with same dimensions. Figure 21 shows the geometric image of Design 2.

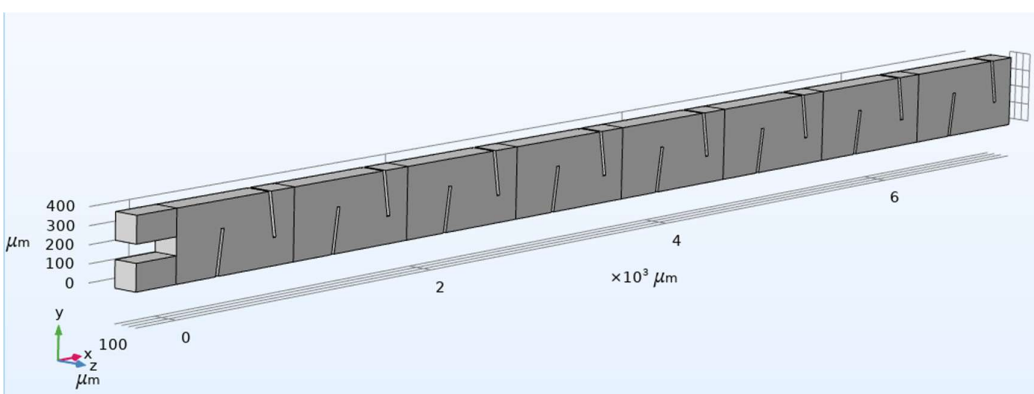

Figure 21: Design 2 Geometry with 8 Mixing Units

Because the geometry is in such microscopic scales the meshing of a longer geometry was difficult, therefore the geometry was divided into smaller components were created and addition of all those results can be connected. For this geometry a model was created in COMSOL Multiphysics.

5.3.3.2. Model Setup

Initial Boundary Condition:

At inlet the boundary condition of a fully developed flow with a flow rate 0.1 ml/min were set as the optimal results were for the following flow rate was obtained in base design results. Similar to earlier boundary condition, a fully developed flow condition was set at inlet and zero pressure condition at outlet. Similarly, a no backflow constrain was set to avoid any kind of reluctance to the flow.

In total 3 studies were run for this design. First, a basic stationary study was set to obtain general value of concentration profile and MI was done. Also, to select an optimal mesh quality an adaptive meshing setting was turned on, providing accurate results around edges of the paralleogram barriers. Second, a test to check mesh dependency of the design, a parametric sweep of different mesh size was performed to generate mesh dependent data of the design. Finally, we needed to check the time dependent solution of the concentration and verify the time constrain that it took to reach a certain mixing quality. The time range for which the results were generated was taken 0.5 seconds with time steps for 0.05 seconds. For all these studies a GMRES solver was used, and the residual error limit was taken as 0.001 for accurate results.

For this model because the geometry is more complex than earlier it cost most computing power. To mitigate this issue, both laminar solver and transport of dilute species is solved in different study steps. The benefit of this approach is that the peak memory usage of the compter is decreased compared to a combined study which is desirable.

5.3.4. Result Obtained

5.3.4.1 Concentration Profile:

which are part of liquid flow cell Earlier the focus was to obtain concentration profile and velocity profile were generated. The result for the time dependent study shows the following concentration maps at different time steps (Figure 22). The concentration profile at time 0 shows no flow, and when the fluid is allowed to flow inside the channel a zigzag motion of both fluid is seen. This kind of motion is responsible for generating vertical advection component. Therefore, the concentration gradient is becoming more homogeneous as the fluid passes through each pair of barriers in the chamber. The time step result for this study was taken for maximum time of 0.5 seconds. The aim for

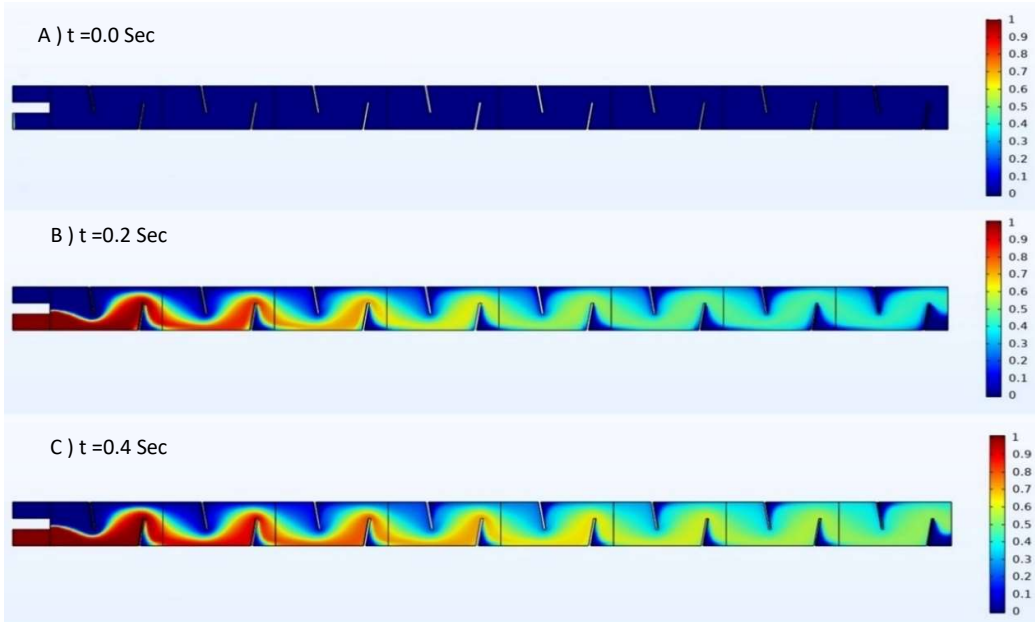

Figure 22 Concentration Map of Design 2 generated through time-dependent COMSOL study for different time steps A) 0 sec, B) 0.2 sec, C 0.4 sec)

time-dependent study was to check if the flow parameters is consistent with time. Data obtained for MI in the tube is plotted with time in Figure 23. Here it highlights that the result for the mixing index at the outlet increases with each time step but becomes constant after 0.3 seconds time has passed. This is the mixing time for the third design.

Now in initial studies the geometry was built for a single pair of parellogram barrier to check effect of single pair of obstacles to the mixing performance of the device. Further studies included of increasing the number of mixing units to find an optimal number. From the studies' data in the form of a table was generated for the mixing with different number of mixing units at time step of 0.4 seconds was generated. Form the data it is clear that after 8 mixing unit, mixing performance is greatly increased 91 % (0.911) compared to that of single unit with 41%(0.417) Mixing qualitty.

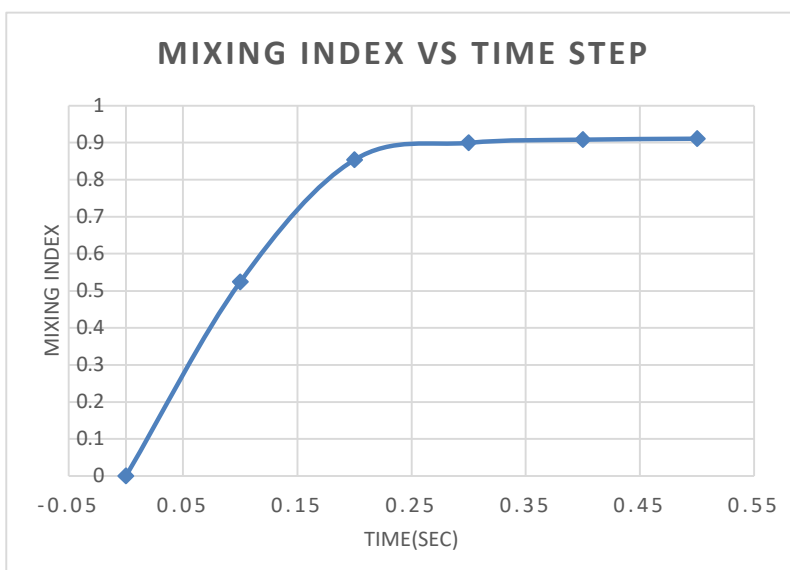

Figure 23: Graph plot of time versus mixing index at outlet for design 2.

Further addition of mixing units provides very minor increase, thus mixing channel with 8 pair of barrier units was found to be the optimal design.

5.3.4.2 Pressure Drop:

While designing micromixer it is necessary to consider the pressure drop occurring in the flow as the device, in practice works as a smaller part for a lab-on-chip (LOC)

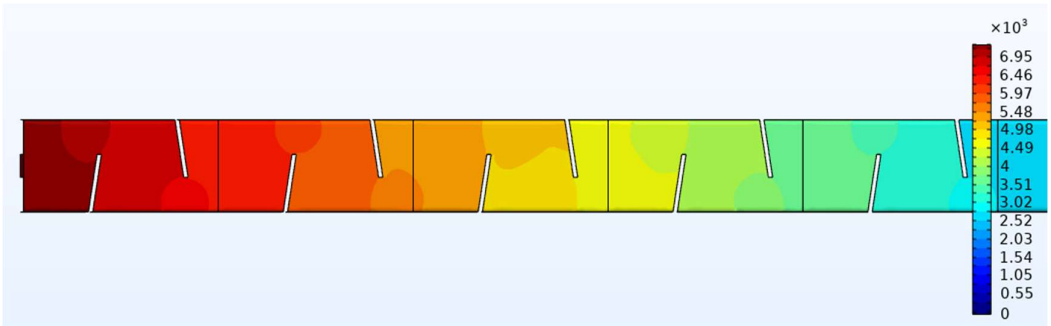

Figure 25: Pressure drop contour of Design 2 .

system (Bhagat and Papautsky, 2008). The pressure drops in this device needed to be calculated in order to calculated for each the barriers and know how much pressure drop is occurring. Pressure drops and mixing efficiency shows an inverse correlation in the plot however lower pressure drop were achieved for negligible change in mixing efficiency by changing the angle of the barriers.

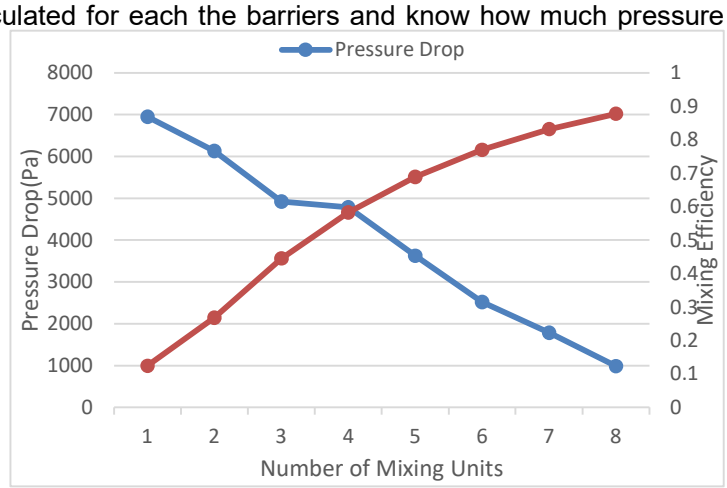

Figure 24: Variation of MI and Pressure drop with variation in number of Mixing units

5.3.4.3 Concentration Cavity:

From the concentration plot it was discovered that by changing the angle of parallelogram barrier the pressure drop can be controlled. However, when barriers are placed with low angles too the horizontal cavity formation occurs. As shown in the Figure 26 locus of concentration just behind the barriers are seen which causes demixing to occur. And this concentration gradient also decreases the advection component in the flow.

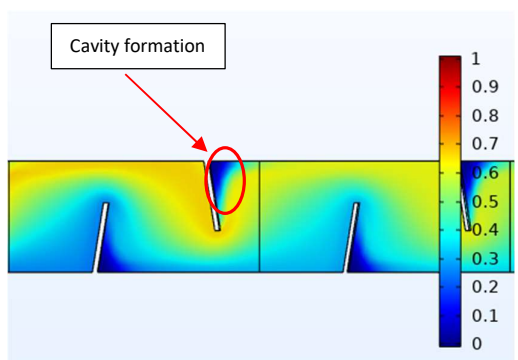

Figure 26 Concentration profile for design 3, showing cavity formation.

Therefore, an optimal angle where high mixing and low pressure drop occurs with minimum concentration cavity formation occurs was calculated. The optimal angle for these barriers was found to be 80.12 degree from the horizontal.

5.3.3.4. Mesh Dependency Test

Similarly to the last design a mesh dependency test by increasing the grid density results in a non-converging result therefore, convergence of non dependent result parameter like velocity flux with the degree of freedoms solved by the solver is generated with the results provided in the Appendix(Section 5). Although the results from the velocity magnitude with respect to the arc length of the micromixer was generated, a zoomed in image is shown below to understant the variation caused by the mesh upon the non dependent velocity magnitude parameter. For reference the plot of velocity magnitude with respect to the while length for different mesh sizes is plovided in the appendix.

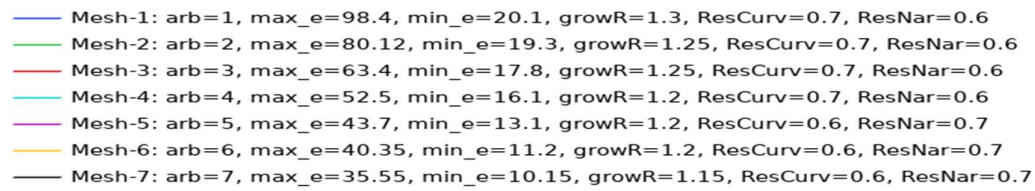

Figure 27: Legend to represent each mesh with the associated line shown in figure 30.

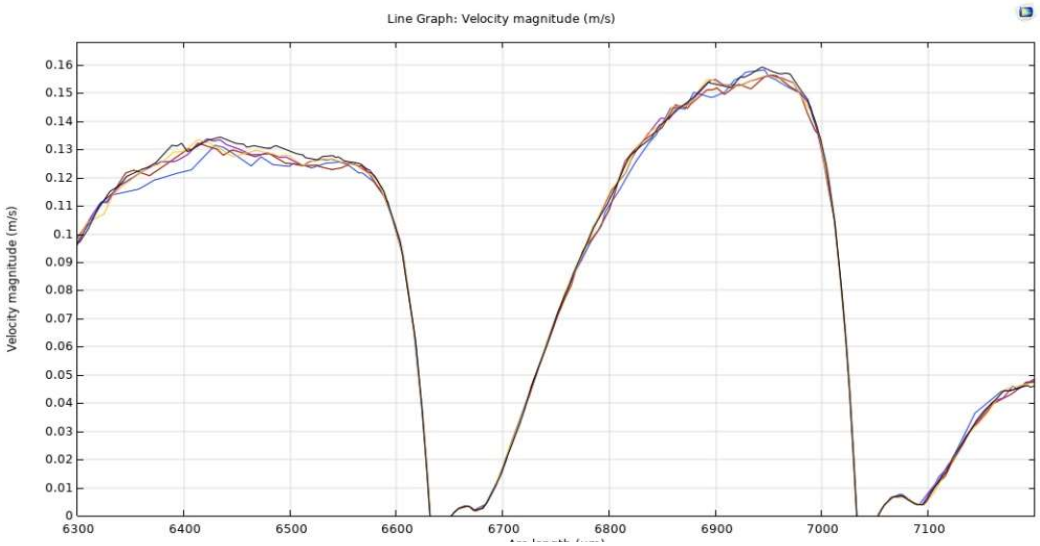

Figure 28: Velocity convergence test result for single micromixer unit near the outlet.

The figure 28 shows the velocity magnitude plot for the arc length at centre of the channel for the length of one micromixer unit for 7 different meshes from coarse to fine. The graph has certain certain points where the velocity is 0 those are the obstructions in the geometry. The figure shows two peaks where there is small difference in velocity however for finer meshes the value of velocity is following one arc that is closest to the finest mesh. Therefore the result provides that after mesh 3 the value of the velocity is

converging with only a slight change in magnitue. Therefore we can say the model is appropriate and further increase in mesh size provides a negligible change in results.

5.4. Design Comparison

All the models computed provides a idea of the performance of the devices. As the base design did not reach the mark of the best performance in mixing aspect design 1 and Design 2 are the competitors for the title. However from the results genrated it can be said that both the designs have some advantages and flaws in different scenarios.

Speaking for Design 1, although the mixing obtained is better that the base design it will be lesser favoured compared to design compared for Mixing Index. It is evident that the concentration disparity is higher of both edges for design 1 which can cause output flow to be considered less homogeneously mixed. Howerer the time it takes for mixing to occur is lower as the obstructions althoug hcauseing the flow to produce advection, still deos not slow down the flow velocity to a big extent, compared to the design 2 where the ZigZag- cyclic flux pathlines are formed which inherently decreases the flow speed.

The pressure drop in design 1 is also lower than design 2, which indicates that lower energy will be required to drive the flow throug hthe chamber which makes it better for some specific drug transport studies where inlet flow is a important study parameter. Therefore selection of the better design will depend upon the requriement of specific studies that are to be conducted.

Chapter 6. Conclusion

6.1. Achievements

The Project report provides various Achievements and results, some of the are described as below.

- Better understanding of Internal fluid structure and interaction properties and inside a flow cell chamber is developed.
- An effect of demixing is confirmed dependent on the geometry and layout of the advection-chaotic flow.
- It was confirmed that introduction of obstructive flow path causes better mixing however causing relative pressure drop.
- Mixing performance data and variation is achieve dfor microfluidic flows.
- Sucessfully geometrical design optimization was achieved to improve mixing performance of already inoperation design of Hummingbird Scientific.

6.2. Discussion

By reference of literature review regarding the flow cell design and mixing chamber designs, various implementation of the micromixers designs were analyzed. Because, Implementation of microfluidics for TEM based studies, two models seems appropriate. As discussed in section 5.4 where it was clear that both design provide certain advantages and for each design geometrical modification increases the mixing result. Therefore modification by geometrical modification is reliable. Although the concern here is the affect this causes in the study design. From design 2 concentration plot it shows.

Because the model was build in COMSOL multiphysics that works on finite element analysis, mesh dependency has poor results therefore similar model needs to be confirmed in other software packages like Ansys. Still convergence of mesh was seen that means the model still predicts correct values however accuracy can be slightly lower relative to a model working on finite volume method.

Also there are more designs which employ micromixers that provides better mixing value for smaller micromixer length. The designs build were specifically modified

design suggestions of the original designs of Hummingbrid Scientific. Thus we can not say that the mixing we obtained here through this study is the most optimal value generated in the whole industry.

The main objective of the project was to understand the flow pattern and interactions which is sucessful and tims report provide firm understanding regarding that, with design suggetion and validative proof.

6.3. Conclusion

In Conclusion, this project obtains a thorough analysis of the flow patterns and mixing performance for various inlet and boundary scenarios. The proposed model implementation on optimized design are found to be superior in terms of mixing quality. A Mixing index of 91 percent and 78 percent of design 1 and design 2 was achieved, which is 2 times and 1.7 time to that of the base design, respectively. Verifiaction study by comparing the result patterns with different research materials and resultant convergence plots prooves the reliability of the implenented model. On the basis of the model the report concludeds and presents two updated designs for further analysis as a prospective design as a micromixer component in the flow cell unit of Hummingbird Scientific.

6.4. Future Work

In the presented report computational analysis was performed for the device. A experimental study for the same geometrical and boundary conditions is suggested to check the error quotient for humar or instrumental error consideration needs to be performed. With that said, the analysis done in this project is purely computational therefore to develop the proposed design into a prospective product multiple studies regarding the structural analysis, interation with different conditions in place like (electromagnetic field, etc) needs to be checked. Finally regarding the computational model, further designs can be built with pulsating flow conditions with different phase difference which increases the mixing drastically.

References

1. Microfluidic Mixers: A short review. 2022. Elverflow. [Online]. [Accessed 15/02/2022]. Available from: "https://www.elveflow.com/microfluidic-reviews/microfluidic-flow-control/microfluidic-mixers-a-short-review/".
2. Hummingbird Scientific 2015 . [Online]. [Accessed 10/01/2022]. Available from: "https://hummingbirdscientific.com/wp-content/uploads/Liquid-Holder-WEB-Brochure.pdf".
3. Y. C. Tan, J. S. Fisher, A. I. Lee, V. Cristini, and A. P. Lee, "Design of microfluidic channel geometries for the control of droplet volume, chemical concentration, and sorting," Lab on a Chip 4. 292-298 (2004).
4. Hummingbird Scientific. 2022. [Online]. [Accessed 15 February 2022]. Available from: "https://hummingbirdscientific.com/products/liquid/#".
5. H*ummingbird Scientific*.2022. [Online]. [Accessed 15 February 2022]. Available from: "https://hummingbirdscientific.com/sem-liquid-overview/".
6. Schwesinger, N., Frank, T. and Wurmus, H. 1996. A modular microfluid system with an integrated micromixer. *Journal of Micromechanics and Microengineering*. 6(1),pp.99-102.
7. The transport of diluted species interface documentation,. COMSOL Multiphysics®. 2021. pp. 75-84 [Online]. [Accessed 18/05/2022]. Available from: "https://doc.comsol.com/5.6/doc/com.comsol.help.chem/chem_ug_chemsptrans.08.049.html#:~:text=The%20Transport%20of%20Diluted%20Species%20(tds)%20interface%20(%20)%2C,be%20handled%20with%20this%20interface".
8. Cengel, Y.A, and Cimbala, J.M, 2017. Fluid Mechanics: Fundamentals and Applications(4th edition). McGraw-Hill Education.
9. Douglas J. Pike, Nikhil Kapur, Paul A. Millner, and Douglas I. Stewart. 2013. Flow Cell Design for Effective Biosensing. Sensors. 13(1),pp.58-70.
10. Liangchao Z, Lanru J, Vladimir C. Modelling of flow and mixing in 3D rough-walled rock fracture intersections. 2017. Advances in Water Resources. 107. pp.1-9.
11. Lee C, Chang C, Wang Y, and Fu L. Microfluidic mixing: a Review. 2011. Interantional Journal of Molecular Sciences. 12(5). pp.3263-3287.
12. Lee, C., Wang, W., Liu, C. and Fu, L., 2016. Passive mixers in microfluidics systems: A review. Chemical Engineering Journal. 288. Pp. 146-160.
13. Thabet, S. and Thabit, T. 2018. Computational fluid Dynamics: Science of the Future. International Journal of Research and Engineering. 5(6). Pp 430-433.
14. Itomlenskis, M., Fodor, P. and Kaufman, M., 2008. Design of Passive Micromixers using the COMSOL Multiphysics Software Package.
15. Mondal Bappa, Sukumar Pati & Pramod Kumar Patowari. 2019. Analysis of mixing performances in microchannel with obstacles of different aspect ratios. Proceedings of

the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering. 233(). pp. 1045-1051.

16. Mengeaud V, Girault HH, Josserand J. Mixing processes in a zigzag microchannel: finite element simulations and optical study. Anal Chem. 2002. pp. 4279-4286.
17. Ali Asgar S Bhagat, Sanjay Bhagat & Papautsky Ian. 2008. Enhancing particle dispersion in a passive planar micromixer using rectangular obstacles. Journal of Micromechanical and Microengineering. 18(8). p. 85005-85009.
18. Nguyen N and Wu Z. 2005. Micromixers- A review. Journal of Micromechanical and Microengineering. 15(2). pp. R1-R16.
19. Frei, W. 2013. Solutions to Linear Systems of Equations: Direct and Iterative Solvers. *COMSOL*. [Online]. [Accessed 01/08/2022]. Available from: "https://www.comsol.com/blogs/solutions-linear-systems-equations-direct-iterative-solvers/#:~:text=The%20direct%20solvers%20used%20by,some%20quite%20ill%2Dconditioned%20problems".
20. Williams, M., Longmuir, K. and Yager, P., 2008. A practical guide to the staggered herringbone mixer. *Lab on a Chip*, 8(7), p.1121.
21. R. Prichard, M. Gibson, C. Joseph, W. Strasser. 2022. A review of fluid flow in and around the brain, modeling, and abnormalities. Multiscale Biomechanical Modeling of the Brain, Academic Press. pp. 209-238.
22. Park, G., Kim, G., Han, D., Kim, K., Shin, S., Park, J. and Park, Y. 2017. Visualization and label-free quantification of microfluidic mixing using quantitative phase imaging. Applied Optics, 56(22), p.6341.

Appendix:

Section 5:

Design 1

1)

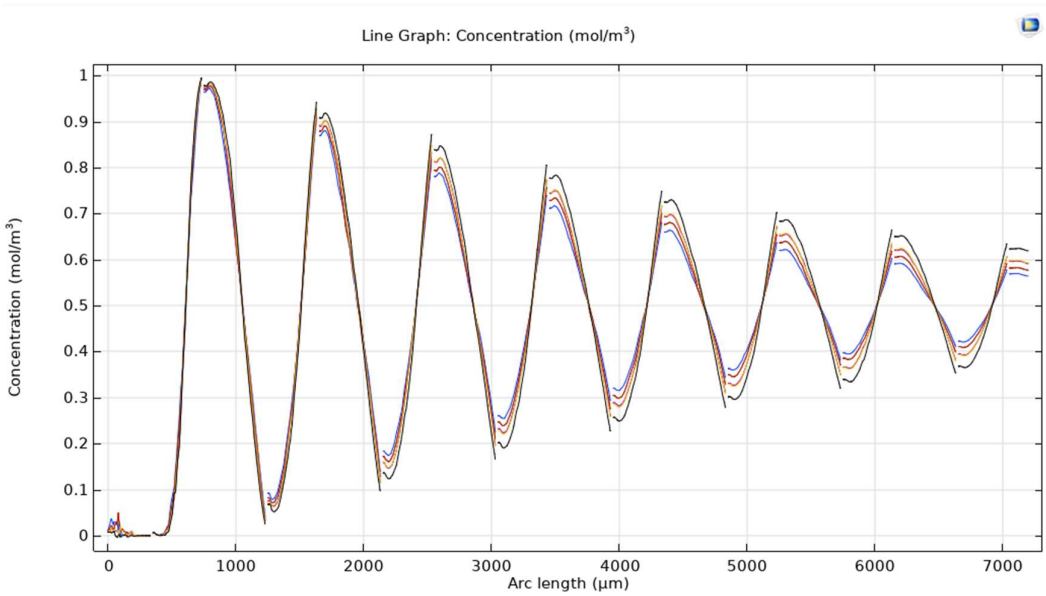

Figure 29: Mesh Convergence study for design 1(Full Graph)

2) Parametric study result of design 1 showing table for different value for different mesh.

Mesh Number	Max size	Min Size	Growth	Curvature	Narrow resolution	Internal DOF	DOF	Mixing Index	Mixing Flux Index
1	98.4	20.1	1.4	0.8	0.5	34278	482767	0.74515	0.74967
2	80.12	19.3	1.35	0.7	0.6	34579	506187	0.72949	0.73241
3	63.4	17.8	1.3	0.7	0.6	34625	529492	0.71259	0.71446
4	52.5	16.1	1.25	0.7	0.6	34679	560452	0.6911	0.69264
5	43.7	13.1	1.2	0.6	0.7	35234	624117	0.66439	0.66448
6	40.35	11.2	1.17	0.6	0.7	39364	732187	0.64336	0.64135
7	35.55	10.15	1.15	0.6	0.6	41590	829117	0.62	0.61973

3) Time Dependent study result of design 2 for each time steps.

Time	Mixing Index(Outlet)	Mixing flux index(Outlet)	MI at position 1	MI at position 2	MI at position 3	MI at position 4	MI at position 5	MI at position 6	MI at position 7
0	0	0	1.50E-08	9.99E-16	3.33E-16	1.11E-15	3.33E-16	7.77E-16	8.88E-16
0.1	0.52389	0.22575	8.99E-09	0.34987	0.55179	0.64563	0.64847	0.60023	0.52653
0.2	0.85398	0.44354	4.64E-09	0.30409	0.50385	0.64882	0.74414	0.79187	0.79983
0.3	0.89983	0.52927	3.32E-09	0.28336	0.47162	0.61564	0.72396	0.80051	0.8498
0.4	0.90839	0.58954	2.53E-09	0.27324	0.45457	0.59478	0.7027	0.78328	0.84356
0.5	0.91065	0.63717	1.97E-09	0.26785	0.44542	0.58297	0.68924	0.76955	0.83185

Design 2:

1)

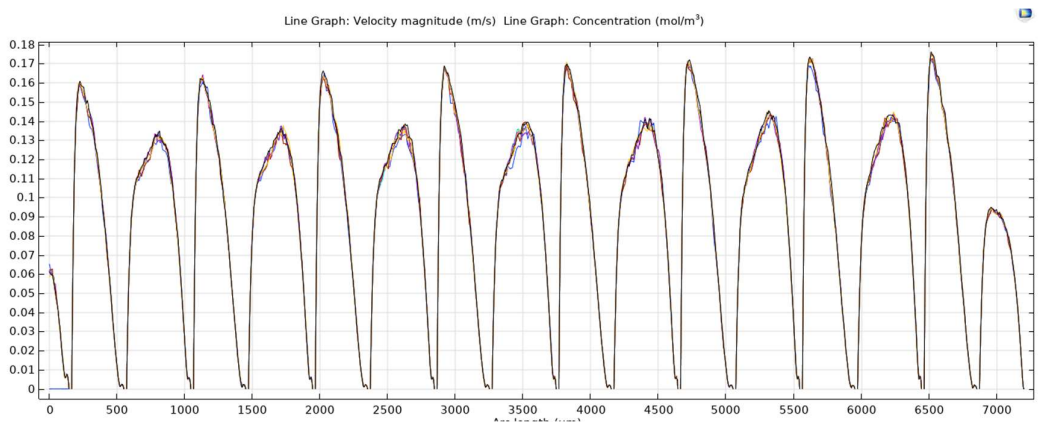

Figure 30: Mesh Convergence study for design 2(Full Graph)

2) Parametric study result of design 2 showing table for different value for different mesh.

Mesh Number	Max size	Min Size	Growth	Curvature	Narrow resolution	Internal DOF	DOF	Mixing Index	Mixing Flux Index
1	98.4	20.1	1.3	0.7	0.6	28711	495692	0.88494	0.92175
2	80.12	19.3	1.25	0.7	0.6	28854	523322	0.86404	0.90735
3	63.4	17.8	1.25	0.7	0.6	28854	523322	0.86404	0.90735
4	52.5	16.1	1.2	0.7	0.6	28885	562717	0.83844	0.88811
5	43.7	13.1	1.2	0.6	0.7	28885	562767	0.83888	0.88832
6	40.35	11.2	1.2	0.6	0.7	28870	563582	0.83648	0.88672
7	35.55	10.15	1.15	0.6	0.7	28978	632172	0.7921	0.85202

3) Time Dependent study result of design 2 for each time steps.

Time	Mixing Index(Outlet)	Mixing flux index(Outlet)	MI at position 1	MI at position 2	MI at position 3	MI at position 4	MI at position 5	MI at position 6	MI at position 7
0	0	0	1.50E-08	3.05E-06	-6.60E-11	-4.44E-16	1.11E-16	-6.66E-16	0
0.05	1.06E-04	1.59E-04	1.50E-08	0.067283	0.23716	0.34011	0.41681	0.44227	0.39208
0.1	0.37527	0.41616	5.85E-09	0.058752	0.23325	0.33729	0.42468	0.4957	0.5536
0.15	0.70315	0.7133	1.32E-09	0.058651	0.23304	0.33737	0.42557	0.49851	0.56136
0.2	0.74865	0.75083	9.48E-10	0.058665	0.23302	0.33736	0.42565	0.49879	0.56213
0.25	0.75413	0.7554	9.07E-10	0.058661	0.23301	0.33737	0.42566	0.49882	0.56222
0.3	0.75457	0.75581	9.04E-10	0.058661	0.23301	0.33737	0.42566	0.49882	0.56223